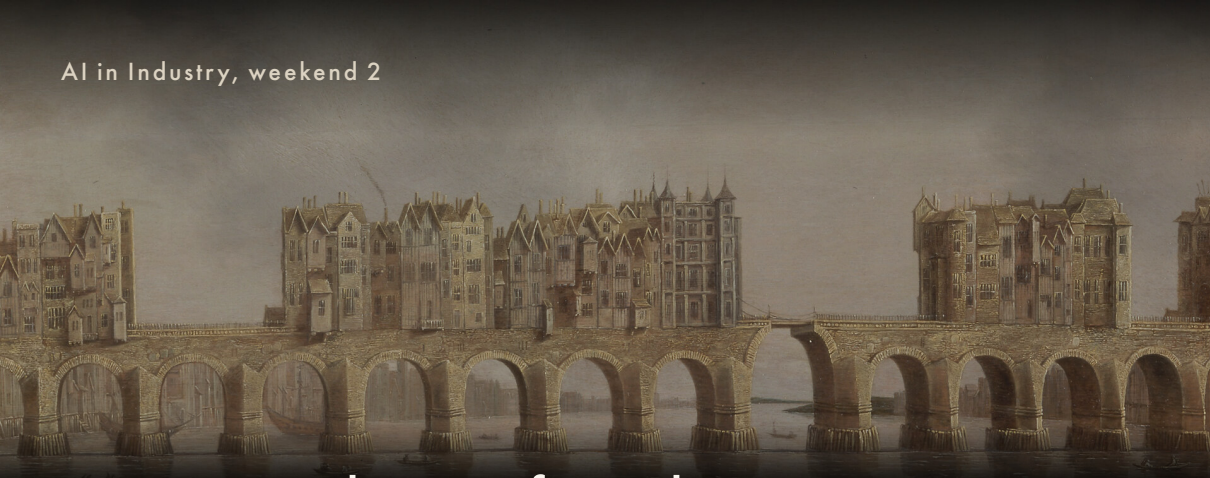
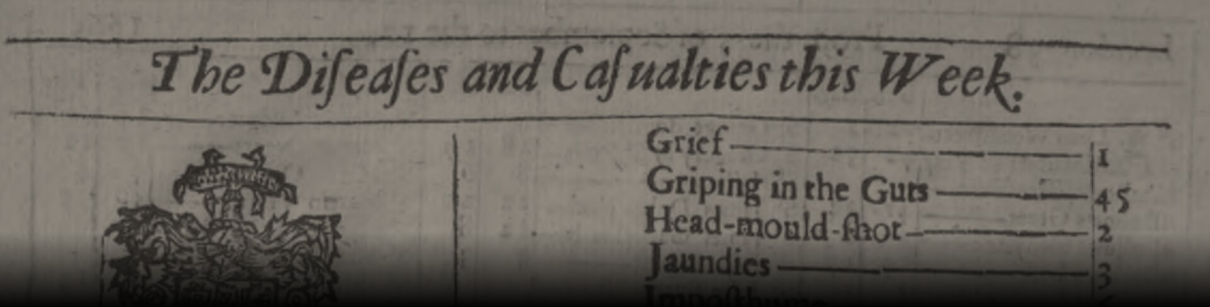
AI in Industry, weekend 2



Estimating chances from the past

A book from London, 1662

Claude de Jongh, *View of London Bridge*, 1632, detail. Yale Center for British Art. Wikimedia Commons, public domain

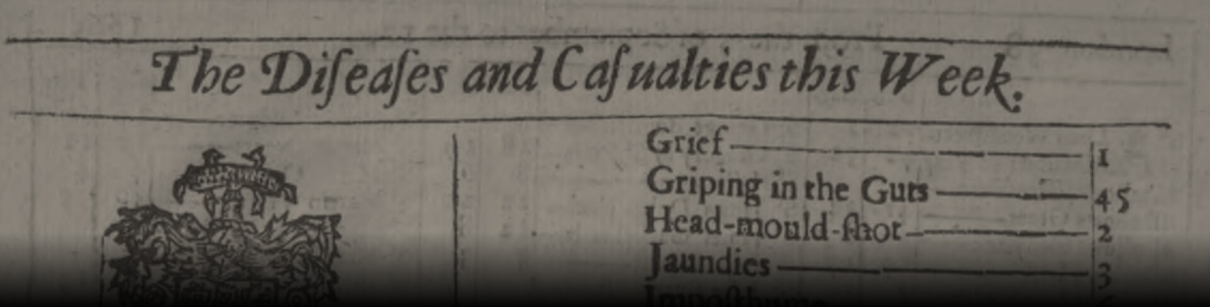


The Bills of Mortality

London, seventeenth century

Every week, the parish clerks of London print the burials of the week and what each person died of. Londoners read these bills in fear of the plague and of many other diseases.

Parish Clerks of London, bill of 5 to 12 September 1665, detail.
US National Library of Medicine. Internet Archive, public domain



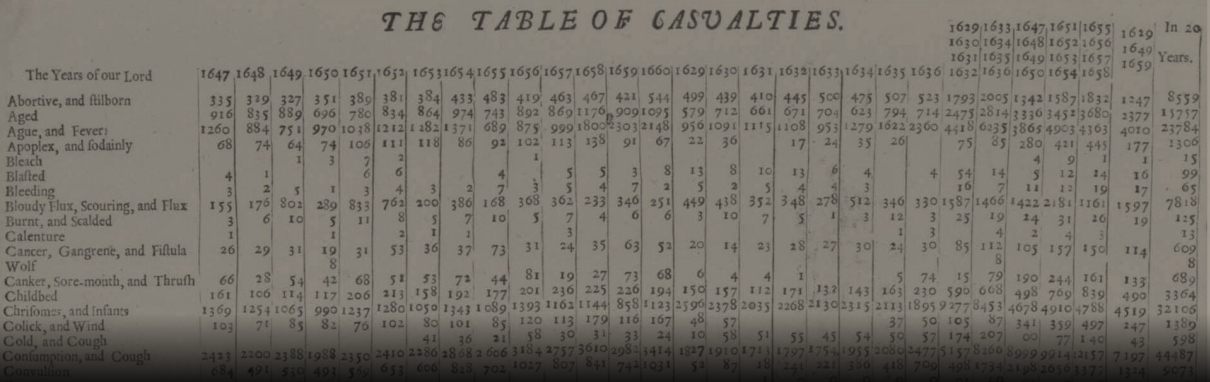
The Bills of Mortality

London, seventeenth century

Every week, the parish clerks of London print the burials of the week and what each person died of. Londoners read these bills in fear of the plague and of many other diseases.

How afraid should a Londoner be of each disease?

Parish Clerks of London, bill of 5 to 12 September 1665, detail.
US National Library of Medicine. Internet Archive, public domain


THE TABLE OF CASUALTIES.

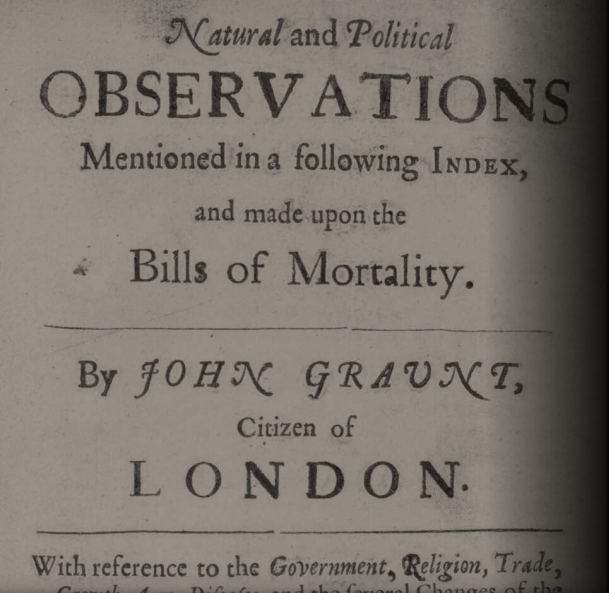
The Years of our Lord	1647	1648	1649	1650	1651	1652	1653	1654	1655	1656	1657	1658	1659	1660	1629	1630	1631	1632	1633	1634	1635	1636	1629 1630 1631 1632	1633 1634 1635 1636	1647 1648 1649 1650	1651 1652 1653 1654	1655 1656 1657 1658	1629 1649 1659	In 20 Years.
Abortive, and ſtilborn	335	329	327	351	389	381	384	433	483	419	463	467	421	544	499	439	410	445	500	475	507	523	1793	2005	1342	1587	1832	1247	8559
Aged	916	835	889	696	780	834	864	974	743	892	869	1176	909	1095	579	712	661	671	704	623	794	714	2475	2814	3336	3452	3680	2377	15757
Ague, and Feveri	1260	884	751	970	1038	1212	1182	1371	689	875	999	1800	2303	2148	956	1091	1115	1108	953	1279	1622	2360	4418	6235	3865	4903	4363	4010	23784
Apoplex, and ſodainly	68	74	64	74	106	111	118	86	92	102	113	158	91	67	22	36		17	24	35	26		75	85	280	421	445	177	1306
Bleach			1	3	7	2				1															4	9	1	1	15
Blaſted	4	1			6	6			4		5	5	3	8	13	8	10	13	6	4		4	54	14	5	12	14	16	99
Bleeding	3	2	5	1	3	4	3	2	7	3	5	4	7	2	5	2	5	4	4	3			16	7	11	12	19	17	65
Bloudy Flux, Scouring, and Flux	155	176	802	289	833	762	200	386	168	368	362	233	346	251	449	438	352	348	278	512	346	330	1587	1466	1422	2181	1161	1597	7818
Burnt, and Scalded	3	6	10	5	11	8	5	7	10	5	7	4	6	6	3	10	7	5	1	3	12	3	25	19	24	31	26	19	125
Calenture	1			1		2	1	1			3										1	3		4	2	4	3		13
Cancer, Gangrene, and Fiſtula	26	29	31	19	31	53	36	37	73	31	24	35	63	52	20	14	23	28	27	30	24	30	85	112	105	157	150	114	609
Wolf				8																				8					8
Canker, Sore-mouth, and Thruſh	66	28	54	42	68	51	53	72	44	81	19	27	73	68	6	4	4	1		5	74	15	79	190	244	161	133	689
Childbed	161	106	114	117	206	213	158	192	177	201	236	225	226	194	150	157	112	171	132	143	163	230	590	668	498	769	839	490	3364
Chriſoms, and Infants	1369	1254	1065	990	1237	1280	1050	1343	1089	1393	1162	1144	858	1123	2596	2378	2035	2208	2130	2315	2113	1895	9277	8453	4078	4910	4788	4519	32106
Colick, and Wind	103	71	85	82	76	102	80	101	85	120	113	179	116	167	48	57					37	50	105	87	341	359	497	247	1389
Cold, and Cough						41	36	21	58	30	31	33	24	10	58	51	55	45	54	50	57	174	207	60	77	140	43	598
Conſumption, and Cough	2423	2200	2388	1988	2350	2410	2280	2808	2666	3184	2757	3610	2982	3414	1827	1910	1713	1797	1754	1955	2080	2477	5157	8200	8999	9914	12137	7197	44487
Convulſion	684	491	510	493	569	653	600	828	702	1027	807	841	742	1031	51	87	18	241	221	330	418	709	498	1734	2198	2656	3377	1314	9071

I shall onely set down how many died of each.

John Graunt, a haberdasher of London, 1662

John Graunt, *Natural and Political Observations... upon the Bills of Mortality*, London, 1662, chapter 2

John Graunt, *Natural and Political Observations*, 1662, the Table of Casualties, detail. Wikimedia Commons, public domain


Natural and *Political*

OBSERVATIONS

Mentioned in a following INDEX,

and made upon the

Bills of Mortality.

By JOHN GRAUNT,

Citizen of

LONDON.

With reference to the *Government*, *Religion*, *Trade*, *Growth*, *Ayre*, *Diseases*, and the several Changes of the

John Graunt, *Natural and Political Observations*, 1662, title page, detail. Wikimedia Commons, public domain

John Graunt

1620 to 1674, haberdasher

"Now having (I know not by what accident) engaged my thoughts upon the Bills of Mortality, and so far succeeded therein, as to have reduced several great confused Volumes into a few perspicuous Tables…"

John Graunt, *Natural and Political Observations... upon the Bills of Mortality*, London, 1662, the dedication

Of 229250 deaths

twenty years of bills

Drowned	829
Sodainly	454
Grief	279
Lunatique	158
Murthered	86
Head-Ach	51
Frighted	22
Excessive drinking	2

"That not one in two thousand are Murthered in London."

John Graunt, *Natural and Political Observations... upon the Bills of Mortality*, London, 1662, chapter 2 and the index

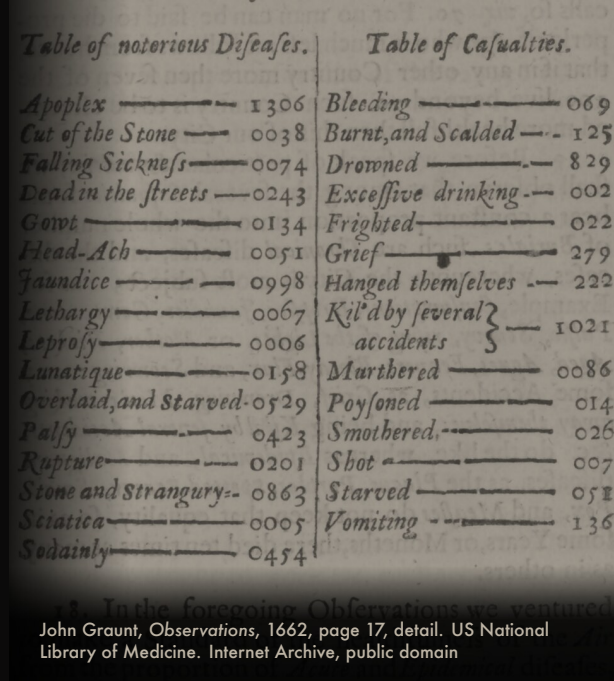


Table of noterious Diseases.

Apoplex	1306
Cut of the Stone	0038
Falling Sickness	0074
Dead in the streets	0243
Gowt	0134
Head-Ach	0051
Jaundice	0998
Lethargy	0067
Leprosy	0006
Lunatique	0158
Overlaid, and Starved	0529
Palsy	0423
Rupture	0201
Stone and Strangury	0863
Sciatica	0005
Sodainly	0454

Table of Casualties.

Bleeding	069
Burnt, and Scalded	125
Drowned	829
Excessive drinking	002
Frighted	022
Grief	279
Hanged themselves	222
Kil'd by several accidents	1021
Murthered	0086
Poysoned	014
Smothered	026
Shot	007
Starved	051
Vomiting	136

13. In the foregoing Observations we ventured

John Graunt, *Observations*, 1662, page 17, detail. US National Library of Medicine. Internet Archive, public domain



"Whereas many persons live in great fear, and apprehension of some of the more formidable, and notorious diseases... I shall onely set down how many died of each: that the respective numbers, being compared with the Total 229250, those persons may the better understand the hazard they are in."

John Graunt, *Natural and Political Observations... upon the Bills of Mortality*, London, 1662, chapter 2

Claude de Jongh, *View of London Bridge*, 1632, detail. Yale Center for British Art. Wikimedia Commons, public domain



"Whereas many persons live in great fear, and apprehension of some of the more formidable, and notorious diseases... I shall onely set down how many died of each: that the respective numbers, being compared with the Total 229250, those persons may the better understand the hazard they are in."

Graunt divides each count by all the deaths.
From the past, every Londoner can estimate the hazard of each disease.

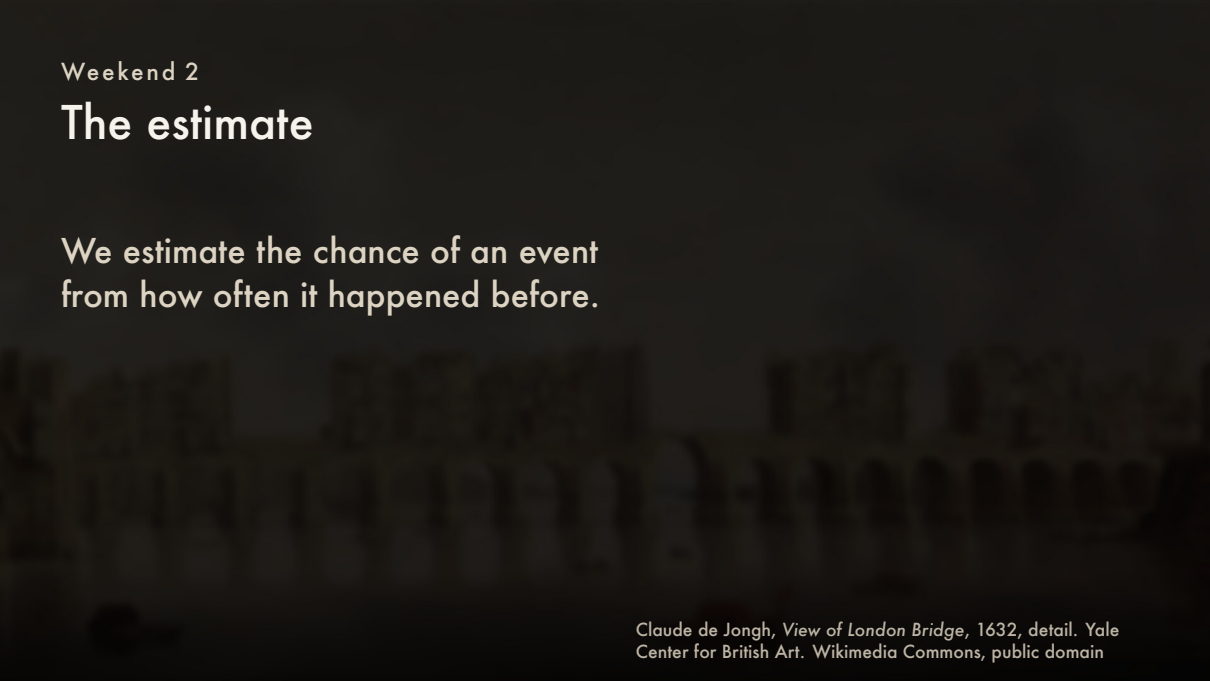
John Graunt, *Natural and Political Observations... upon the Bills of Mortality*, London, 1662, chapter 2

Claude de Jongh, *View of London Bridge*, 1632, detail. Yale Center for British Art. Wikimedia Commons, public domain

Weekend 2

The estimate

We estimate the chance of an event from how often it happened before.



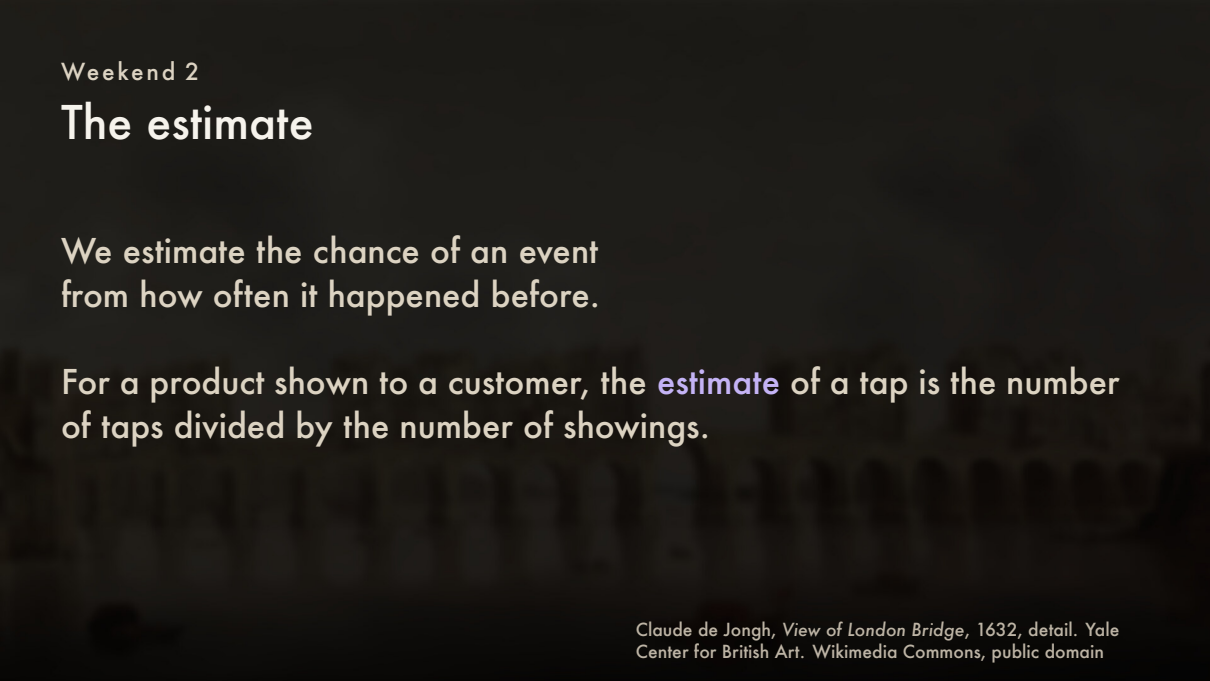
Claude de Jongh, *View of London Bridge*, 1632, detail. Yale Center for British Art. Wikimedia Commons, public domain

Weekend 2

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For a product shown to a customer, the estimate of a tap is the number of taps divided by the number of showings.



Claude de Jongh, *View of London Bridge*, 1632, detail. Yale Center for British Art. Wikimedia Commons, public domain

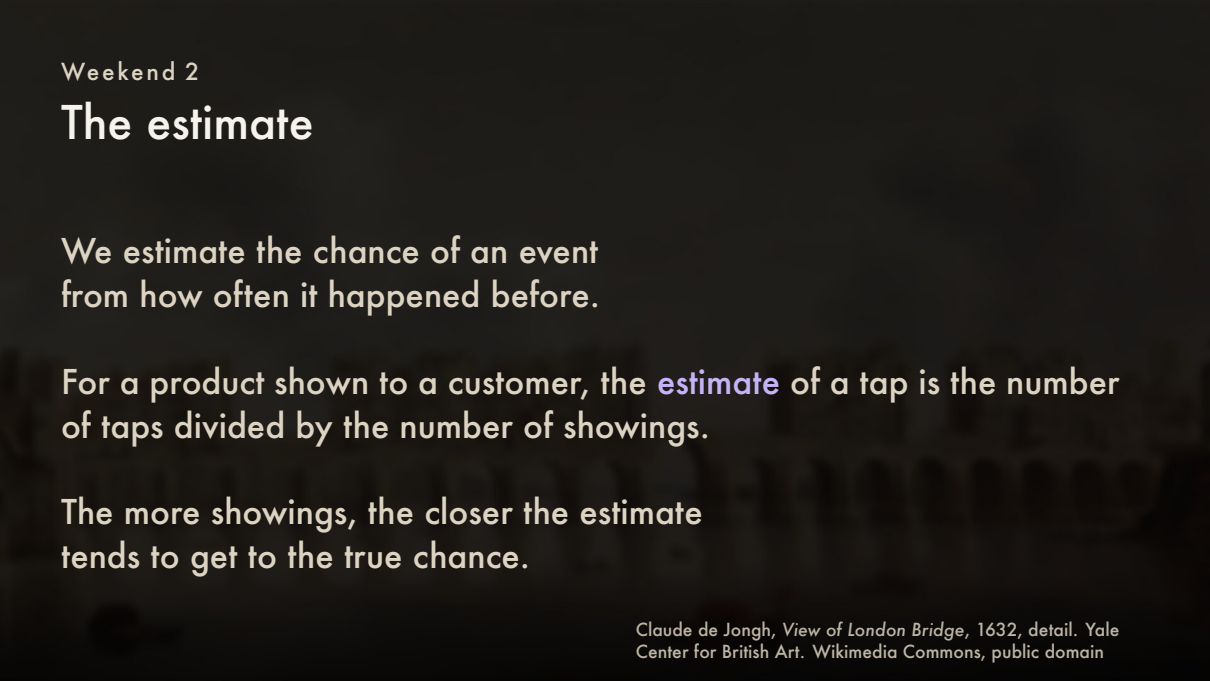
Weekend 2

The estimate

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The more showings, the closer the estimate
tends to get to the true chance.



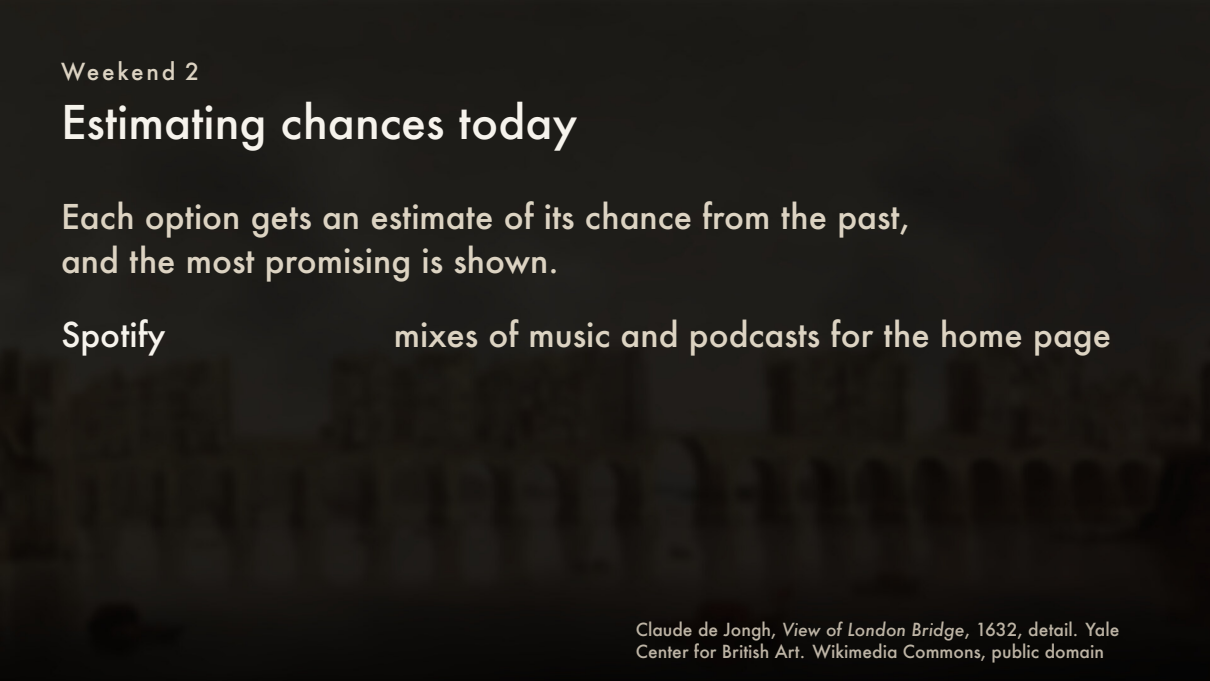
Claude de Jongh, *View of London Bridge*, 1632, detail. Yale Center for British Art. Wikimedia Commons, public domain

Weekend 2

Estimating chances today

Each option gets an estimate of its chance from the past,
and the most promising is shown.

Spotify mixes of music and podcasts for the home page



Claude de Jongh, *View of London Bridge*, 1632, detail. Yale Center for British Art. Wikimedia Commons, public domain

Weekend 2

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Spotify	mixes of music and podcasts for the home page
Amazon Payments	images and taglines for a page

Claude de Jongh, *View of London Bridge*, 1632, detail. Yale Center for British Art. Wikimedia Commons, public domain

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Such estimates decide much of what we are shown online.

Claude de Jongh, *View of London Bridge*, 1632, detail. Yale Center for British Art. Wikimedia Commons, public domain

Business cases

Bandits

Carlos Cotrini
AI in Industry, ETH Zurich
Friday 13 November 2026

1

Section 1

Spotify's home page

Spotify's home page: music or podcasts?

The home page shows shelves of music and podcasts. A common rule gives users the mix of their past listening.

But tastes can change with the hour, the day of the week, even the device.

Feijer and others (Spotify), CONSEQUENCES workshop at RecSys 2025.

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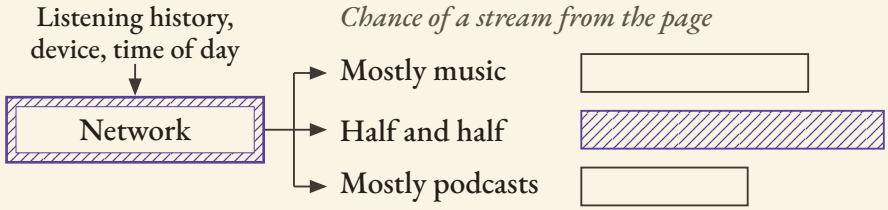
Over 600 million

users of the home page

Feijer and others (Spotify), CONSEQUENCES workshop at RecSys 2025.

Epsilon-greedy on the home page

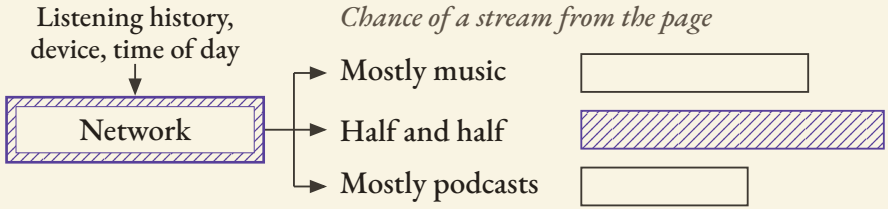
The mirror counted taps. Here a network estimates the chances.



Invented mixes and estimates. *Feijer and others (Spotify), CONSEQUENCES workshop at RecSys 2025.*

Epsilon-greedy on the home page

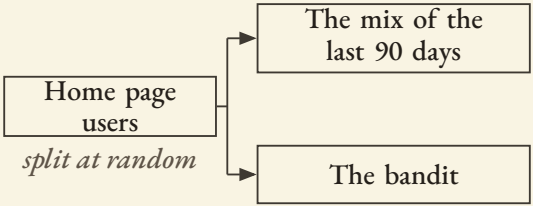
The mirror counted taps. Here a network estimates the chances.



98.5% of the pages: the highest estimate.
1.5%: a random mix near the usual one.

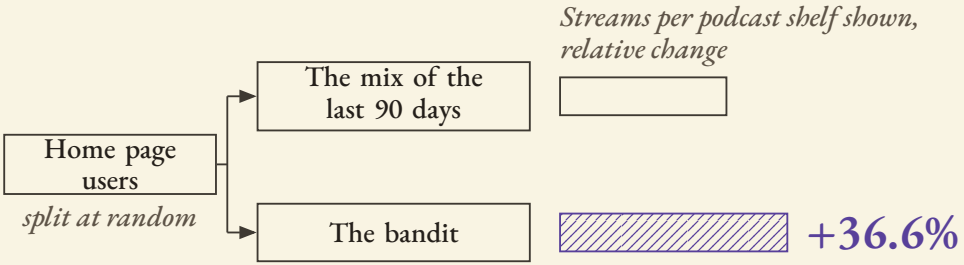
Invented mixes and estimates. *Feijer and others (Spotify), CONSEQUENCES workshop at RecSys 2025.*

The value: an A/B test



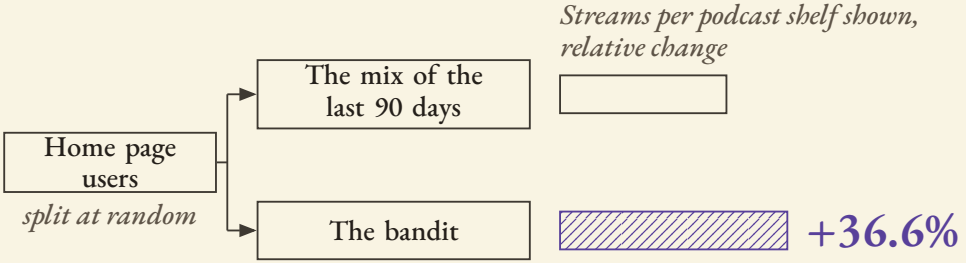
Feijer and others (Spotify), CONSEQUENCES workshop at RecSys 2025.

The value: an A/B test



Feijer and others (Spotify), CONSEQUENCES workshop at RecSys 2025.

The value: an A/B test



In production as of March 2025.

+1.28% *minutes listened on Spotify, relative change*

Feijer and others (Spotify), CONSEQUENCES workshop at RecSys 2025.

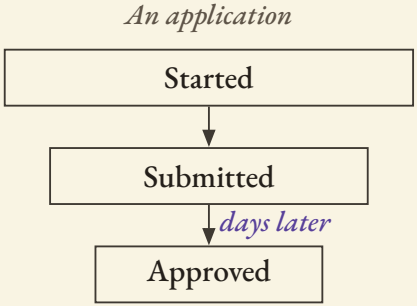
2

Section 2

Amazon Payments

Amazon Payments: which page for which visitor?

A page invites customers to apply for an Amazon Payments product. It shows one image and one tagline, from many possible pairs.

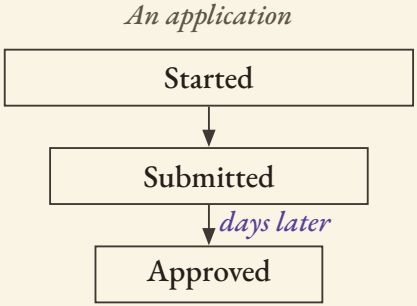


John and Gozluklu (Amazon), AWS Machine Learning Blog, 2026.

Amazon Payments: which page for which visitor?

A page invites customers to apply for an Amazon Payments product. It shows one image and one tagline, from many possible pairs.

A page that wins the first step can lose the last.



John and Gozluklu (Amazon), AWS Machine Learning Blog, 2026.

One UCB score for each step

Each score is an estimate plus a bonus, as in the mirror.

For one visitor	*Started*	*Submitted*	*Approved*
Pair A	6	3	1
Pair B	4	4	4
Pair C	3	2	3

The first step alone would choose pair A.

Invented scores. *John and Gozluklu (Amazon), AWS Machine Learning Blog, 2026.*

One UCB score for each step

Each score is an estimate plus a bonus, as in the mirror.

For one visitor	*Started*	*Submitted*	*Approved*	*Sum*
Pair A	6	3	1	10
Pair B	4	4	4	12
Pair C	3	2	3	8

The first step alone would choose pair A.
Amazon shows the highest sum: pair B.

Invented scores. John and Gozluklu (Amazon), AWS Machine Learning Blog, 2026.

The value: an A/B test of 7 weeks

The bandit against the fixed page:

One group of customers



Approvals rose by a high single-digit percentage.
relative change so far, not reported as significant

John and Gozluklu (Amazon), AWS Machine Learning Blog, 2026.

The value: an A/B test of 7 weeks

The bandit against the fixed page:

One group of customers		Approvals rose by a high single-digit percentage. *relative change so far, not reported as significant*
Another group		No pair it tried beat the fixed page. Approvals fell. *statistically significant*

John and Gozluklu (Amazon), AWS Machine Learning Blog, 2026.

The value: an A/B test of 7 weeks

The bandit against the fixed page:

One group of customers		Approvals rose by a high single-digit percentage. *relative change so far, not reported as significant*
Another group		No pair it tried beat the fixed page. Approvals fell. *statistically significant*

Amazon: “Thorough exploration can confirm that a content pool contains no winner.”

John and Gozluklu (Amazon), AWS Machine Learning Blog, 2026.